

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“JnanaSangama”, Belgaum -590014, Karnataka.



LAB REPORT on

Object Oriented Java Programming (23CS3PCOOJ)

Submitted by

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in partial fulfillment for the award of the degree of
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CERTIFICATE

This is to certify that the Lab work entitled “Object Oriented Java Programming (23CS3PCOOJ)” carried out by **Shashank Shinde(1BF24CS277)**, who is bonafide student of **B.M.S. College of Engineering**. It is in partial fulfilment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum. The Lab report has been approved as it satisfies the academic requirements in respect of an Object-Oriented Java Programming (23CS3PCOOJ) work prescribed for the said degree.

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Program 1:

Implent Quadratic Equation:

CODE:

```
import java.util.*;
import java.lang.Math;
public class quadratic {
    public static void main
(String args[])
    {
        Scanner sc = new
Scanner (System.in);
        double r1,r2;
```

```
System.out.println("Ente
r the value of a, b, c:");
        int a=sc.nextInt();
        int b=sc.nextInt();
        int c=sc.nextInt();
        if(a==0)
        {
            while(a==0){
```

```
System.out.println("Not
a quadratic equation.");
```

```
System.out.println("Ente
r a non zero value of a:");
        a=sc.nextInt();
        }
    }
```

```

        double d=b*b-
(4*a*c);
        if(d==0)
        {
            r1=(-b)/(2*a);
            r2=r1;

```

```

System.out.println("Both
the roots are real and
equal.");

```

```

System.out.println("The
roots are r1 =
"+r1+"\t"+"r2 = "+r2);
        }
        else if(d>0)
        {
            r1=(((-b)+Math.sqrt(d))/(double)
(2*a));
            r2=(((-b)-
Math.sqrt(d))/(double)(2*
a));

```

```

System.out.println("The
roots are real and
unequal.");

```

```

System.out.println("The
roots are r1 =
"+r1+"\t"+"r2 = "+r2);
        }
        else
        {

```

```

        r1=(-
b)/(double)(2*a);
        r2=Math.sqrt(-
d)/(double)(2*a);

```

```

System.out.println("The
roots are imaginary and
unequal.");

```

```

System.out.println("the
roots are r1 =
"+r1+"+i("+r2+")\t"+r2 =
"+r1+"-i("+r2+")");
    }
}
}

```

```

PS C:\Users\Shashank Shinde\Desktop\java programs> cd "c:\Users\Shashank Shinde\Desktop\java programs\" ; if ($?) { ja
vac quadratic.java } ; if ($?) { java quadratic }
Enter the value of a, b, c:
1
-4
4
Both the roots are real and equal.
The roots are r1 = 2.0 r2 = 2.0
PS C:\Users\Shashank Shinde\Desktop\java programs> cd "c:\Users\Shashank Shinde\Desktop\java programs\" ; if ($?) { ja
vac quadratic.java } ; if ($?) { java quadratic }
Enter the value of a, b, c:
1
-4
12
The roots are imaginary and unequal.
the roots are r1 = 2.0+i(2.8284271247461903) r2 = 2.0-i(2.8284271247461903)
PS C:\Users\Shashank Shinde\Desktop\java programs> cd "c:\Users\Shashank Shinde\Desktop\java programs\" ; if ($?) { ja
vac quadratic.java } ; if ($?) { java quadratic }
Enter the value of a, b, c:
1
-4
-12
The roots are real and unequal.
The roots are r1 = 6.0 r2 = -2.0
PS C:\Users\Shashank Shinde\Desktop\java programs>

```

Program 2:

Compute SGPA :

CODE:

```
import java.util.Scanner;
class Subject{
    int
    subjectMarks,credit,grade;
}
public class student {
    String name,usn;
    double sgpa;
    Scanner sc = new
Scanner(System.in);
    Subject subject[];
    student(int n)
    {
        subject = new
Subject[n];
        for(int i=0;i<n;i++)
        {
            subject[i]=new
Subject();
        }
    }
    void
getStudentDetails()
    {

System.out.println("Enter student name:");
```

```

        name= sc.next();

System.out.println("Enter USN:");
        usn=sc.next();
    }
    void getMarks(int n)
    {
        for(int i=0;i<n;i++)
        {

System.out.println("Enter subject "+(i+1)+"
marks :");

        subject[i].subjectMarks=
sc.nextInt();

System.out.println("Enter subject "+(i+1)+"
credits :");

        subject[i].credit=sc.nextInt();

        subject[i].grade=(subject
[i].subjectMarks/10)+1;

        if(subject[i].grade>=11)
        {

        subject[i].grade = 10;
        }

```



```

        else
if(subject[i].grade<=4)
    {

subject[i].grade=0;
    }
    }
    }
    double
computeSGPA(int n)
    {
        int totalcredits=0;
        int
totalgrade=0;
        for(int i=0;i<n;i++)
        {

totalcredits=totalcredits+
subject[i].credit;

totalgrade=totalgrade+
subject[i].grade;
        }
        double sgpa=0.0;

sgpa=(double)totalgrade/
totalcredits;
        return sgpa;
    }
    void display(int n)
    {

```

```
System.out.println("Name  
of student : "+name);
```

```
System.out.println("USN  
: "+usn);
```

```
System.out.println("SGP  
A : "+computeSGPA(n));  
}
```

```
public static void  
main(String args[])  
{  
    Scanner s= new  
Scanner(System.in);
```

```
System.out.println("Enter  
the number of subjects :  
");
```

```
    int n=s.nextInt();  
    student st = new  
student(n);
```

```
st.getStudentDetails();  
    st.getMarks(n);
```

```
st.computeSGPA(n);  
    st.display(n);  
    s.close();
```

```
}
```

```
}
```

```
Enter the details of student:
Enter name of student:
shashank
Enter the usn of the student:
123
Enter the number of subjects:
4
Enter marks of1 subject
87
Enter credits of1 subject
3
Enter marks of2 subject
92
Enter credits of2 subject
3
Enter marks of3 subject
87
Enter credits of3 subject
3
Enter marks of4 subject
89
Enter credits of4 subject
3
Student details:
Student name:shashank
Student usn:123
Student sgpa:9.0
```

Program 3:

Implement toString():

CODE:

```
import java.util.*;
class Books{
    String name,author;
    int price,num_pages;
    Books(String
name,String author,int
price,int num_pages)
    {
        this.name=name;
        this.author=author;
        this.price=price;

this.num_pages=num_p
ages;
    }
    public String toString()
    {
        String
name,author,price,num_
pages;
        name="Book name :
"+this.name+"\n";
        author="Author
name :
"+this.author+"\n";
        price="Price :
"+this.price+"\n";

num_pages="Number of
```

```

Pages :
"+this.num_pages+" \n";
    return
name+author+price+nu
m_pages;
}
}
public class MainBooks
{
    public static void
main(String[] args)
    {
        Scanner sc = new
Scanner(System.in);
        String name,author;
        int
n,price,num_pages;

System.out.println("Ente
r the number of books: ");
        n=sc.nextInt();
        Books b[];
        b= new Books[n];

System.out.println("Ente
r details of the books : ");
        for(int i=0;i<n;i++)
        {

System.out.println("Ente
r name of the book
"+(i+1)+" : ");
            name=sc.next();

```

```
System.out.println("Enter name of the author : ");
    author=sc.next();
```

```
System.out.println("Enter the price : ");
    price=sc.nextInt();
```

```
System.out.println("Enter the number of pages : ");
```

```
num_pages=sc.nextInt();
    b[i]=new
Books(name,author,price,num_pages);
```

```
}
```

```
for(int i=0;i<n;i++)
{
```

```
System.out.println("\nDetails of the book "+(i+1)+" :");
```

```
System.out.println(b[i].toString());
```

```
}
```

```
}
```

}

```
Enter the number of books:
2
Enter details of the books :
Enter name of the book 1 :
harry
Enter name of the author :
potter
Enter the price :
400
Enter the number of pages :
800
Enter name of the book 2 :
hamlet
Enter name of the author :
shakespeare
Enter the price :
500
Enter the number of pages :
1500

Details of the book 1 :
Book name : harry
Author name : potter
Price : 400
Number of Pages : 800

Details of the book 2 :
Book name : hamlet
Author name : shakespeare
Price : 500
Number of Pages : 1500

PS C:\Users\Lenovo\Desktop\3rd sem java>
```

Program 4:

Abstract Area:

CODE:

```
import java.util.Scanner;
abstract class Shape
{
    int a,b;
    Shape(int a, int b)
    {
        this.a=a;
        this.b=b;
    }
    abstract void
    printArea();

}
class Rectangle extends
Shape
{
    Rectangle(int l,int b)
    {
        super(l,b);

    }
    void printArea()
    {

        System.out.println("Area
        of Rectangle =" +(a*b));
    }

}
class Triangle extends
Shape
{
```



```

    Triangle(int height,int
base)
    {
        super(height,base);
    }
    void printArea()
    {

```

```

System.out.println("Area
of Triangle =
"+(0.5*a*b));
    }
}

```

```

class Circle extends
Shape
{
    Circle(int r)
    {
        super(r,0);
    }
    void printArea()
    {

```

```

System.out.println("Area
of Circle = "+(3.14*a*a));

    }
}

```

```

public class
AbstractArea
{
    public static void
main(String args[])

```

```

    {
        int n1,n2;
        Scanner sc= new
Scanner(System.in);

System.out.println("Enter the dimensions of
rectangle(length,
breadth) :");
        n1=sc.nextInt();
        n2=sc.nextInt();
        Rectangle rec=
new Rectangle(n1,n2);
        rec.printArea();

System.out.println("Enter the dimensions of
Triangle(height, base) :");
        n1=sc.nextInt();
        n2=sc.nextInt();
        Triangle tri =new
Triangle(n1,n2);
        tri.printArea();

System.out.println("Enter the dimension of
circle(radius) :");
        n1=sc.nextInt();
        Circle cir=new
Circle(n1);
        cir.printArea();
    }
}

```

```
Enter the dimensions of rectangle(length, breadth) :  
4  
5  
Area of Rectangle =20  
Enter the dimensions of Triangle(height, base) :  
5  
5  
Area of Triangle = 12.5  
Enter the dimension of circle(radius) :  
7  
Area of Circle = 153.86  
PS C:\Users\STUDENT\Desktop\1BF24CS277>
```

Program 5:

Banking:

CODE:

```

import java.util.Scanner;

class Bank {
    double balance;
    double rate = 0.05;
}

class SavingsAccount
extends Bank {

    void deposit(int amt) {
        balance += amt;

        System.out.println("Dep
osit successful...");

        System.out.println("Dep
osited : " + amt + "
Updated Balance : " +
balance);
    }

    void withdraw(int amt)
{
    if (amt > balance) {

        System.out.println("Insuf
ficient Balance...");
        return;
    }
    balance -= amt;
}

```

```
System.out.println("With  
drawal successful...");
```

```
System.out.println("With  
drawn : " + amt +  
"\tUpdated Balance : " +  
balance);  
}
```

```
void interest() {
```

```
System.out.println("Rate  
offered by bank : 5%");  
    double interest =  
balance * rate;  
    balance += interest;
```

```
System.out.println("Inter  
est : " + interest +  
"\tUpdated Balance : " +  
balance);  
}
```

```
void display() {
```

```
System.out.println("Acco  
unt Type : Savings");
```

```
System.out.println("Bala  
nce : " + balance);  
}  
}
```

```

class CurrentAccount
extends Bank {

    CurrentAccount() {
        balance = 1000;
    }

    void deposit(int amt) {
        balance += amt;

        System.out.println("Dep
osit successful...");

        System.out.println("Dep
osited : " + amt + "
Updated Balance : " +
balance);
    }

    void withdraw(int amt)
    {
        if (balance < 1000)
        {

            System.out.println("Bala
nce less than Minimum
Balance...");
            int penalty = 500;
            balance -=
penalty;

            System.out.println("Pena
lty imposed : " + penalty);

```

```

        return;
    }

    if (amt > balance) {

System.out.println("Insuf
ficient Balance...");
        return;
    }

    balance -= amt;

System.out.println("With
drawal successful...");

System.out.println("With
drawn : " + amt +
"\tUpdated Balance : " +
balance);
    }

    void interest() {

System.out.println("Inter
est facility not available
for Current Account.");
    }

    void display() {

System.out.println("Acco
unt Type : Current
Account");

```

```
System.out.println("Balance : " + balance);  
    }  
}
```

```
public class banking {  
    public static void  
main(String[] args) {
```

```
        Scanner sc = new  
Scanner(System.in);
```

```
System.out.print("Enter  
customer name : ");  
    String name =  
sc.next();
```

```
System.out.print("Enter  
account number : ");  
    int accno =  
sc.nextInt();
```

```
        SavingsAccount  
sav = new  
SavingsAccount();  
        CurrentAccount cur  
= new CurrentAccount();  
  
        while (true) {
```



```
System.out.println("\n----  
-----MENU-----");
```

```
System.out.println("1.De  
posit");
```

```
System.out.println("2.Wit  
hdraw");
```

```
System.out.println("3.Co  
mpute Interest");
```

```
System.out.println("4.Dis  
play");
```

```
System.out.println("5.Exi  
t");
```

```
System.out.print("Enter  
your choice : ");
```

```
        int ch =  
sc.nextInt();  
        String accty;
```

```
        switch (ch) {
```

```
            case 1:
```

```
System.out.print("Enter  
type of account  
(savings/current) : ");
```

```

        accty =
sc.next();

System.out.print("Enter
amount : ");
        int dAmt =
sc.nextInt();
        if
(accty.equalsIgnoreCase
("savings"))

sav.deposit(dAmt);
        else if
(accty.equalsIgnoreCase
("current"))

cur.deposit(dAmt);
        else

System.out.println("Invalid
account type");
        break;

```

case 2:

```

System.out.print("Enter
type of account
(savings/current) : ");
        accty =
sc.next();

System.out.print("Enter
amount : ");

```

```

        int wAmt =
sc.nextInt();
        if
(accty.equalsIgnoreCase("savings"))

sav.withdraw(wAmt);
        else if
(accty.equalsIgnoreCase("current"))

cur.withdraw(wAmt);
        else

System.out.println("Invalid account type");
        break;

```

case 3:

```

System.out.print("Enter
type of account
(savings/current) : ");
        accty =
sc.next();
        if
(accty.equalsIgnoreCase("savings"))

sav.interest();
        else if
(accty.equalsIgnoreCase("current"))

```

```

cur.interest();
        else

System.out.println("Invalid account type");
        break;

        case 4:

System.out.print("Enter type of account (savings/current) : ");
        accty =
sc.next();

System.out.println("Customer Name : " + name);

System.out.println("Account Number : " + accno);
        if
(accty.equalsIgnoreCase("savings"))

sav.display();
        else if
(accty.equalsIgnoreCase("current"))

cur.display();
        else

```

```
System.out.println("Invalid account type");  
    break;
```

```
    case 5:
```

```
System.exit(0);
```

```
    default:
```

```
System.out.println("Invalid choice");  
    }  
    }  
    }  
}
```

```

PS C:\Users\Shashank Shinde\Desktop\java programs> cd "c:\Users\Shashank Shinde\Desktop\java programs\" ; if ($?) { java banking.java } ; if ($?) { java banking }
Enter customer name : shashank
Enter account number : 12345

-----MENU-----
1.Deposit
2.Withdraw
3.Compute Interest
4.Display
5.Exit
Enter your choice : 1
Enter type of account (savings/current) : savings
Enter amount : 1000
Deposit successful...
Deposited : 1000 Updated Balance : 1000.0

-----MENU-----
1.Deposit
2.Withdraw
3.Compute Interest
4.Display
5.Exit
Enter your choice : 2
Enter type of account (savings/current) : savings
Enter amount : 500
Withdrawal successful...
Withdrawn : 500 Updated Balance : 500.0

-----MENU-----
1.Deposit
2.Withdraw
3.Compute Interest
4.Display
5.Exit
Enter your choice : 3
Enter type of account (savings/current) : savings
Rate offered by bank : 5%
Interest : 25.0 Updated Balance : 525.0

```

```
-----MENU-----
1.Deposit
2.Withdraw
3.Compute Interest
4.Display
5.Exit
Enter your choice : 4
Enter type of account (savings/current) : savings
Customer Name : shashank
Account Number : 12345
Account Type : Savings
Balance : 525.0
```

```
-----MENU-----
1.Deposit
2.Withdraw
3.Compute Interest
4.Display
5.Exit
Enter your choice : 1
Enter type of account (savings/current) : current
Enter amount : 1000
Deposit successful...
Deposited : 1000 Updated Balance : 2000.0
```

```
-----MENU-----
1.Deposit
2.Withdraw
3.Compute Interest
4.Display
5.Exit
Enter your choice : 2
Enter type of account (savings/current) : current
Enter amount : 3000
Insufficient Balance...
```

```
-----MENU-----
1.Deposit
2.Withdraw
3.Compute Interest
4.Display
5.Exit
Enter your choice : 2
Enter type of account (savings/current) : 1500
Enter amount : 1500
Invalid account type

-----MENU-----
1.Deposit
2.Withdraw
3.Compute Interest
4.Display
5.Exit
Enter your choice : 2
Enter type of account (savings/current) : current
Enter amount : 1500
Withdrawal successful...
Withdrawn : 1500      Updated Balance : 500.0

-----MENU-----
1.Deposit
2.Withdraw
3.Compute Interest
4.Display
5.Exit
Enter your choice : 2
Enter type of account (savings/current) : current
Enter amount : 100
Balance less than Minimum Balance...
Penalty imposed : 500
```



```

-----MENU-----
1.Deposit
2.Withdraw
3.Compute Interest
4.Display
5.Exit
Enter your choice : 3
Enter type of account (savings/current) : current
Interest facility not available for Current Account.

-----MENU-----
1.Deposit
2.Withdraw
3.Compute Interest
4.Display
5.Exit
Enter your choice : 4
Enter type of account (savings/current) : current
Customer Name : shashank
Account Number : 12345
Account Type : Current Account
Balance : 0.0

-----MENU-----
1.Deposit
2.Withdraw
3.Compute Interest
4.Display
5.Exit
Enter your choice : 5
PS C:\Users\Shashank Shinde\Desktop\java programs> cd "c:\Users\Shashank Shinde\Desktop\java programs\" ; if ($?) { j
vac banking.java } ; if ($?) { java banking }

```

Program 6:

Implementing

Packages:

CODE:

```
package program6.cie;
```

```
public class Internals  
extends Student {  
    public int[]  
internalMarks = new  
int[5];
```

```
    public Internals(String  
usn, String name, int  
sem, int[] marks) {  
        super(usn, name,  
sem);  
        for (int i = 0; i < 5;  
i++) {  
            internalMarks[i] =  
marks[i];  
        }  
    }  
}
```

```
package program6.cie;
```

```
public class Student {  
    public String usn;  
    public String name;  
    public int sem;
```

```

    public Student(String
usn, String name, int
sem) {
        this.usn = usn;
        this.name = name;
        this.sem = sem;
    }
}

```

```

package program6.see;
import
program6.cie.Student;
public class Externals
extends Student{
    public int[] seeMarks
= new int[5];

```

```

    public Externals
(String usn, String name,
int sem, int[] marks) {
        super(usn, name,
sem);
        for (int i = 0; i < 5;
i++) {
            seeMarks[i] =
marks[i];
        }
    }
}

```

```

package program6;

```

```
import program6.cie.*;
import program6.see.*;
import java.util.Scanner;
public class FinalMarks {
    public static void
    main(String[] args) {
        Scanner sc = new
        Scanner(System.in);
```

```
        System.out.print("Enter
        number of students: ");
        int n = sc.nextInt();
```

```
        Internals[] internals
        = new Internals[n];
        Externals[]
        externals = new
        Externals[n];
```

```
        for (int i = 0; i < n;
        i++) {
            sc.nextLine();
```

```
        System.out.println("\nEnter
        details of student " +
        (i + 1));
```

```
        System.out.print("USN:
        ");
```

```
        String usn =  
sc.nextLine();
```

```
System.out.print("Name:  
");
```

```
        String name =  
sc.nextLine();
```

```
System.out.print("Semester: ");
```

```
        int sem =  
sc.nextInt();
```

```
        int[] imarks =  
new int[5];  
        int[] emarks =  
new int[5];
```

```
System.out.println("Enter Internal Marks (5  
subjects):");
```

```
        for (int j = 0; j < 5;  
j++) {  
            imarks[j] =  
sc.nextInt();  
        }
```

```
System.out.println("Enter SEE Marks (5  
subjects):");
```

```

        for (int j = 0; j < 5;
j++) {
            emarks[j] =
sc.nextInt();
        }

```

```

        internals[i] = new
Internals(usn, name,
sem, imarks);
        externals[i] =
new Externals(usn,
name, sem, emarks);
    }

```

// Display Final
Marks

```

System.out.println("\nFI
NAL MARKS (Internal +
SEE)");

```

```

    for (int i = 0; i < n;
i++) {

```

```

System.out.println("\nUS
N: " + internals[i].usn);

```

```

System.out.println("Nam
e: " + internals[i].name);

```

```

        for (int j = 0; j < 5;
j++) {
            int finalMark =
internals[i].internalMarks

```

```

[j] +
externals[i].seeMarks[j];

System.out.println("Cour
se " + (j + 1) + ": " +
finalMark);
    }
}
}
}
}

```

```

Enter number of students:
1
Enter details for Student 1
Enter USN:
1234
Enter Name:
sam
Enter Semester:
3
Enter internal marks for subject 1
35
Enter internal marks for subject 2
38
Enter internal marks for subject 3
37
Enter internal marks for subject 4
39
Enter internal marks for subject 5
38
Enter SEE marks for subject 1
90
Enter SEE marks for subject 2
98
Enter SEE marks for subject 3
70
Enter SEE marks for subject 4
80
Enter SEE marks for subject 5
65
Final marks of students
Student 1
USN: 1234
Name: sam
Semester: 3
Final marks:
Subject 1: 80
Subject 2: 87
Subject 3: 72
Subject 4: 79
Subject 5: 70

```

Program 7:

Implementing

exceptions:

CODE:

```
import java.util.*;
class WrongAge extends
Exception{
    WrongAge()
    {
        super("Age
Error...");
    }
    WrongAge(String
message)
    {
        super(message);
    }
}
class InputScanner{
    Scanner sc= new
Scanner (System.in);
}
```



```

class Father extends
InputScanner{
    int FatherAge;
    Father() throws
WrongAge
    {

System.out.println("Enter Father's age :");

FatherAge=sc.nextInt();
    if(FatherAge<0)
    {
        throw new
WrongAge("Father's
Age cannot be less than
0...");
    }
    void Display()
    {

System.out.println("Father's age : "+FatherAge);
    }

}
class Son extends
Father{
    int SonAge;
    Son() throws
WrongAge
    {

```

```

        super();

        System.out.println("Enter Son's age :");

        SonAge=sc.nextInt();
        if(SonAge<0)
        {
            throw new
            WrongAge("Son's Age
            cannot be less than 0...");
        }

        if(SonAge>FatherAge)
        {
            throw new
            WrongAge("Son's age
            cannot be greater than
            Father's age...");
        }
        void Display()
        {

        System.out.println("Son's
        age : "+SonAge);
        }

    }
    public class
    FatherSonAge {
        public static void
        main(String[] args) {

```

```

        try {
            Son obj = new
Son();
            obj.Display();
        }
        catch (WrongAge e)
    {

System.out.println("Exce
ption Caught: " +
e.getMessage());
    }
}
}
}
}

```

```

PS C:\Users\Lenovo\Desktop\3rd sem java> & 'C:\Program Files\Eclipse Adoptium\jdk-17.0.17-hot
spot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Lenovo\AppData\Roami
ng\Code\User\workspaceStorage\6144726ddf30280e482fb84602ba8d12\redhat.java\jdt_ws\3rd sem java_69
6f0df2\bin' 'FatherSonAge'
Enter Father's age :
40
Enter Son's age :
20
Son's age : 20
PS C:\Users\Lenovo\Desktop\3rd sem java> ^C
PS C:\Users\Lenovo\Desktop\3rd sem java>
PS C:\Users\Lenovo\Desktop\3rd sem java> c:; cd 'c:\Users\Lenovo\Desktop\3rd sem java'; & 'C:\Pr
ogram Files\Eclipse Adoptium\jdk-17.0.17-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptio
nMessages' '-cp' 'C:\Users\Lenovo\AppData\Roaming\Code\User\workspaceStorage\6144726ddf30280e482f
b84602ba8d12\redhat.java\jdt_ws\3rd sem java_696f0df2\bin' 'FatherSonAge'
Enter Father's age :
20
Enter Son's age :
40
Exception Caught: Son's age cannot be greater than Father's age...
PS C:\Users\Lenovo\Desktop\3rd sem java> ^C
PS C:\Users\Lenovo\Desktop\3rd sem java>
PS C:\Users\Lenovo\Desktop\3rd sem java> c:; cd 'c:\Users\Lenovo\Desktop\3rd sem java'; & 'C:\Pr
ogram Files\Eclipse Adoptium\jdk-17.0.17-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptio
nMessages' '-cp' 'C:\Users\Lenovo\AppData\Roaming\Code\User\workspaceStorage\6144726ddf30280e482f
b84602ba8d12\redhat.java\jdt_ws\3rd sem java_696f0df2\bin' 'FatherSonAge'
Enter Father's age :
-1
Exception Caught: Father's Age cannot be less than 0...
PS C:\Users\Lenovo\Desktop\3rd sem java> ^C
PS C:\Users\Lenovo\Desktop\3rd sem java>
PS C:\Users\Lenovo\Desktop\3rd sem java> c:; cd 'c:\Users\Lenovo\Desktop\3rd sem java'; & 'C:\Pr
ogram Files\Eclipse Adoptium\jdk-17.0.17-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptio
nMessages' '-cp' 'C:\Users\Lenovo\AppData\Roaming\Code\User\workspaceStorage\6144726ddf30280e482f
b84602ba8d12\redhat.java\jdt_ws\3rd sem java_696f0df2\bin' 'FatherSonAge'

```

```
Enter Father's age :  
40  
Enter Son's age :  
-5  
Exception Caught: Son's Age cannot be less than 0...  
PS C:\Users\Lenovo\Desktop\3rd sem java>
```

Program 8:

Multithreading:

CODE:

```
class BMS extends
```

```
Thread{
```

```
    public void run(){
```

```
        for(int i=0;i<5;i++)
```

```
        {
```

```
            System.out.println("BMS  
CE");
```

```
            try{
```

```
                Thread.sleep(10000);
```

```
            }
```

```
        catch(InterruptedException  
on e)
```

```
        {
```

```
            System.out.println(e);
```

```

        }
    }
}
class CSE extends
Thread{
    public void run()
    {
        for(int i=0;i<5;i++)
        {

System.out.println("Com
puter Science");
            try{

Thread.sleep(2000);
                }

catch(InterruptedException
on e)
            {

System.out.println(e);
                }
            }
        }
    }
}
public class Threads{
    public static void
main(String args[]){
        BMS t1 =new
BMS();

```

```
        CSE t2= new  
CSE();  
        t1.start();  
        t2.start();  
    }  
}
```

```
BMSCE  
Computer Science  
Computer Science  
Computer Science  
Computer Science  
Computer Science  
BMSCE  
BMSCE  
BMSCE  
BMSCE  
PS C:\Users\Admin\Desktop\1BF24CS277> 
```

Program 9:

```
import javax.swing.*;  
import java.awt.*;  
import java.awt.event.*;
```

```
class SwingDemo {  
    SwingDemo() {
```

```
        JFrame jfrm = new  
        JFrame("Divider App");  
        jfrm.setSize(275,  
        150);  
        jfrm.setLayout(new  
        FlowLayout());
```

```
        jfrm.setDefaultCloseOpe  
        ration(JFrame.EXIT_ON  
        _CLOSE);
```

```
        JLabel jlab = new  
        JLabel("Enter the divider  
        and dividend:");
```

```
        JTextField ajtf =  
        new JTextField(8);  
        JTextField bjtf =  
        new JTextField(8);
```

```
        JButton button =  
new JButton("Calculate");
```

```
        JLabel err = new  
JLabel();
```

```
        JLabel alab = new  
JLabel();
```

```
        JLabel blab = new  
JLabel();
```

```
        JLabel anslab =  
new JLabel();
```

```
        jfrm.add(err);  
        jfrm.add(jlab);  
        jfrm.add(ajtf);  
        jfrm.add(bjtf);  
        jfrm.add(button);  
        jfrm.add(alab);  
        jfrm.add(blab);  
        jfrm.add(anslab);
```

```
        ActionListener l =  
new ActionListener() {  
        public void  
actionPerformed(Action  
Event evt) {
```

```
        System.out.println("Actio  
n event from a text  
field");  
        }
```



```
};
```

```
ajtf.addActionListener(l);
```

```
bjtf.addActionListener(l);
```

```
button.addActionListener(  
    new ActionListener() {  
        public void  
        actionPerformed(ActionEvent evt) {  
            try {  
                int a =  
Integer.parseInt(ajtf.getText());  
                int b =  
Integer.parseInt(bjtf.getText());  
                int ans = a /  
b;  

```

```
alab.setText("\nA = " +  
a);
```

```
blab.setText("\nB = " +  
b);
```

```
anslab.setText("\nAns =  
" + ans);  
    }
```

```
        catch  
(NumberFormatException  
n e) {
```

```
    alab.setText("");
```

```
    blab.setText("");
```

```
    anslab.setText("");
```

```
    err.setText("Enter Only  
Integers!");  
}
```

```
        catch  
(ArithmeticException e) {
```

```
    alab.setText("");
```

```
    blab.setText("");
```

```
    anslab.setText("");
```

```
    err.setText("B should be  
NON zero!");  
}
```

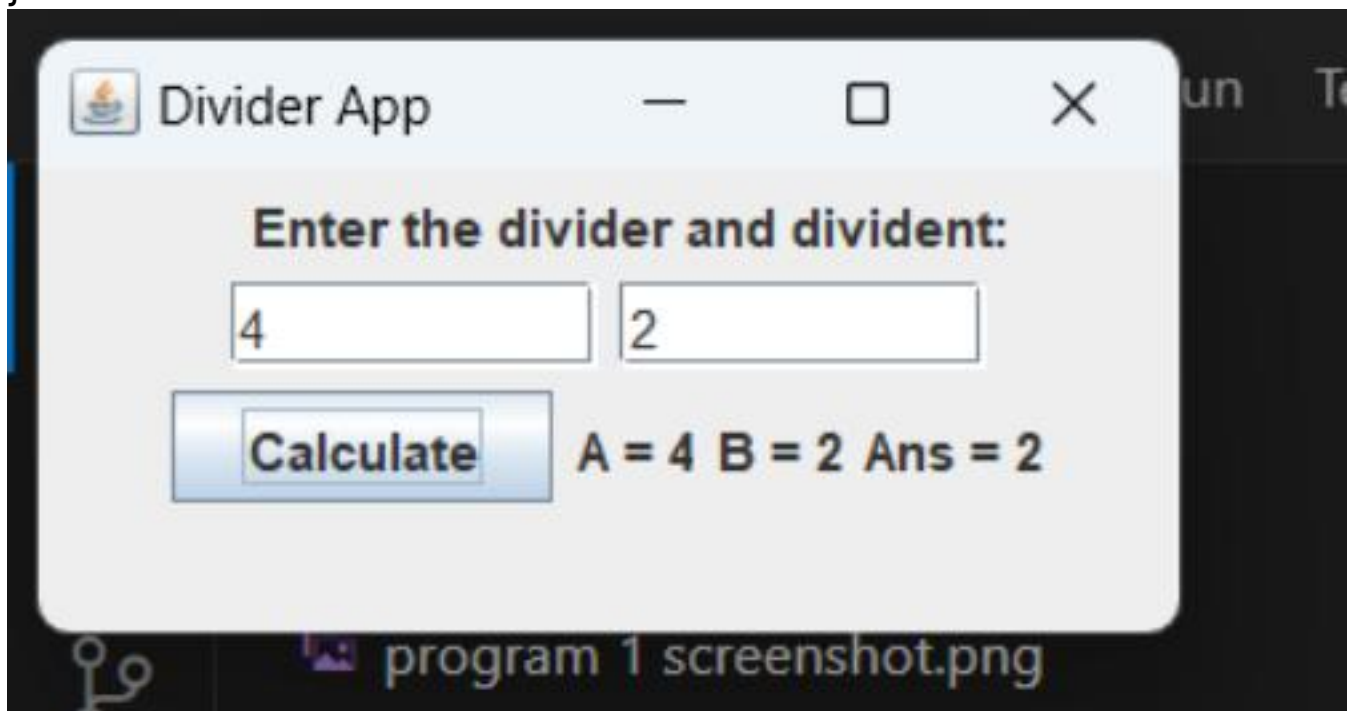
```
    }
```

```
});
```

```
    jfrm.setVisible(true);  
}
```

```
    public static void  
main(String args[]) {
```

```
SwingUtilities.invokeLaterLat  
er(new Runnable() {  
    public void run() {  
        new  
SwingDemo();  
    }  
});  
}  
}
```



Program 10:

```
class Q {
    int n;
    boolean valueSet =
false;

    synchronized int get()
{
    while (!valueSet)
        try {

System.out.println("\nCo
nsumer waiting\n");
            wait();
        } catch
(InterruptedExceptio e)
{

System.out.println("Interr
uptedException caught");
        }

System.out.println("Got:
" + n);
        valueSet = false;

System.out.println("\nInti
mate Producer\n");
        notify();
```

```

        return n;
    }

    synchronized void
    put(int n) {
        while (valueSet)
            try {

                System.out.println("\nPr
                oducer waiting\n");
                wait();
            } catch
            (InterruptedException e)
            {

                System.out.println("Interr
                uptedException caught");
            }

            this.n = n;
            valueSet = true;

            System.out.println("Put:
            " + n);

            System.out.println("\nInti
            mate Consumer\n");
            notify();
        }
    }

    class Producer
    implements Runnable {

```

```
Q q;
```

```
Producer(Q q) {  
    this.q = q;  
    new Thread(this,  
"Producer").start();  
}
```

```
public void run() {  
    int i = 0;  
    while (i < 15) {  
        q.put(i++);  
    }  
}
```

```
class Consumer  
implements Runnable {  
    Q q;
```

```
    Consumer(Q q) {  
        this.q = q;  
        new Thread(this,  
"Consumer").start();  
    }
```

```
    public void run() {  
        int i = 0;  
        while (i < 15) {  
            int r = q.get();
```

```
System.out.println("cons  
umed:" + r);
```

```
        i++;  
    }  
}  
}
```

```
class PCFixed {  
    public static void  
main(String args[]) {  
    Q q = new Q();  
    new Producer(q);  
    new Consumer(q);  
  
    System.out.println("Pres  
s Control-C to stop.");  
    }  
}
```

```

PS C:\Users\Shashank Shinde\Desktop\java programs> cd "c:\Users\Shashank Shinde\Desktop\java programs\" ; if ($?) {
vac PCFixed.java } ; if ($?) { java PCFixed }
Press Control-C to stop.
Put: 0

Intimate Consumer

Producer waiting

Got: 0

Intimate Producer

Put: 1

Intimate Consumer

Producer waiting

consumed:0
Got: 1

Intimate Producer

consumed:1
Put: 2

Intimate Consumer

Producer waiting

Got: 2

Intimate Producer

consumed:2
Put: 3

Intimate Consumer

Producer waiting

Got: 3

Intimate Producer

Put: 4

Intimate Consumer

Producer waiting

consumed:3
Got: 4

Intimate Producer

consumed:4
Put: 5

Intimate Consumer

Producer waiting

Got: 5

Intimate Producer

consumed:5
Put: 6

Intimate Consumer

Producer waiting

```



```
Producer waiting
Got: 10
Intimate Producer
consumed:10
Put: 11
Intimate Consumer

Producer waiting
Got: 11
Intimate Producer
consumed:11
Put: 12
Intimate Consumer

Producer waiting
Got: 12
Intimate Producer
consumed:12
Put: 13
Intimate Consumer

Producer waiting
Got: 13
Intimate Producer
```



```
Got: 6

Intimate Producer

consumed:6
Put: 7

Intimate Consumer

Producer waiting

Got: 7

Intimate Producer

consumed:7
Put: 8

Intimate Consumer

Producer waiting

Got: 8

Intimate Producer

consumed:8
Put: 9

Intimate Consumer

Producer waiting

Got: 9

Intimate Producer

consumed:9
```

```
Intimate Producer

consumed:13
Put: 14

Intimate Consumer

Got: 14

Intimate Producer

consumed:14
PS C:\Users\Shashank Shinde\Desktop\java programs>
```

